Project Asteria - Detailed Design Notes (Review Edition)

These notes summarize the final infrastructure modules designed before implementation.

1. Communication Bus

Acts as the messaging backbone rather than a processing stage. Uses publish-subscribe messaging, message priorities, validation, timestamps, emergency routing, and graceful degradation. Proposed separation into a Data Bus (sensor information) and Command Bus (control commands).

2. Task Scheduler

Allocates CPU time to modules. Supports real-time, high-priority, background, and event-driven tasks. Includes watchdog timers, starvation prevention, adaptive scheduling, and coordination with the Health Monitor.

3. Configuration Manager

Stores all configurable parameters in one location: robot dimensions, sensors, navigation, learning, safety, and mission profiles. Supports runtime updates, versioning, and recommendations for self-tuning while leaving safety-critical changes under human approval.

4. Logging & Debugging Framework

Provides structured logging with DEBUG, INFO, WARNING, ERROR, and CRITICAL levels. Records sensor events, goal transitions, behaviours, failures, timing information, decision explanations, replay capability, black-box recording, and remote monitoring.

5. Updated Architecture

Infrastructure Layer: Communication Bus, Task Scheduler, Configuration Manager, Logging Framework, Health Monitor. Cognitive Pipeline: Sensors → Sensor Fusion → World Model → Memory + Experience Learning → Goal Manager → Behaviour Planner → Motor Controller.

6. Proposed Knowledge Base

Separates long-term facts, rules, mission information, and robot capabilities from the World Model, which represents the current state of the environment.

Next Step

Review every subsystem, simplify where necessary, freeze Asteria Architecture v1.0, then begin Python implementation module-by-module.